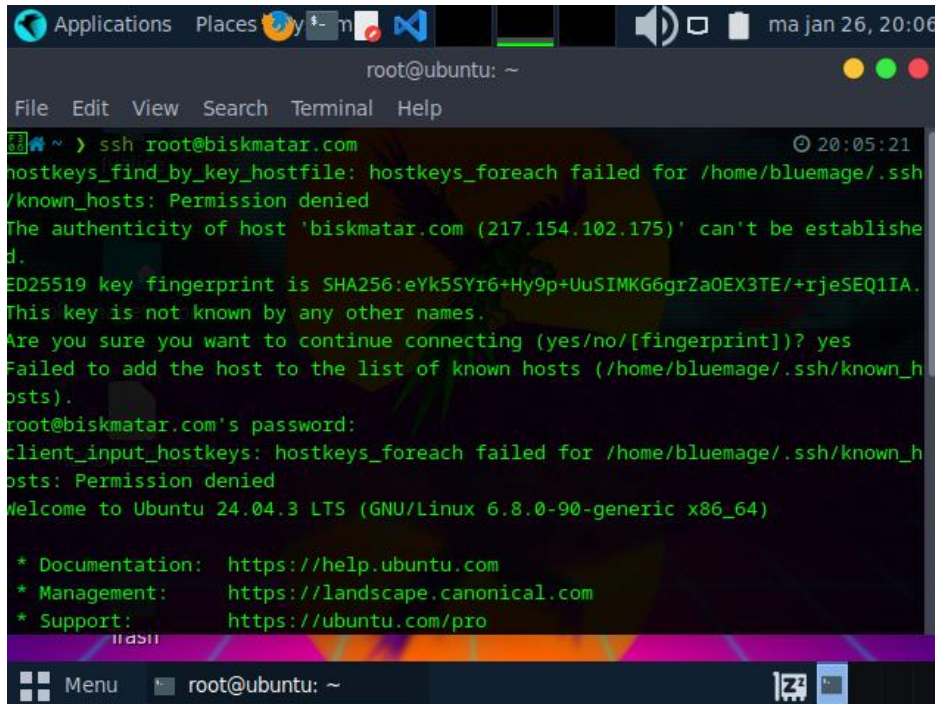


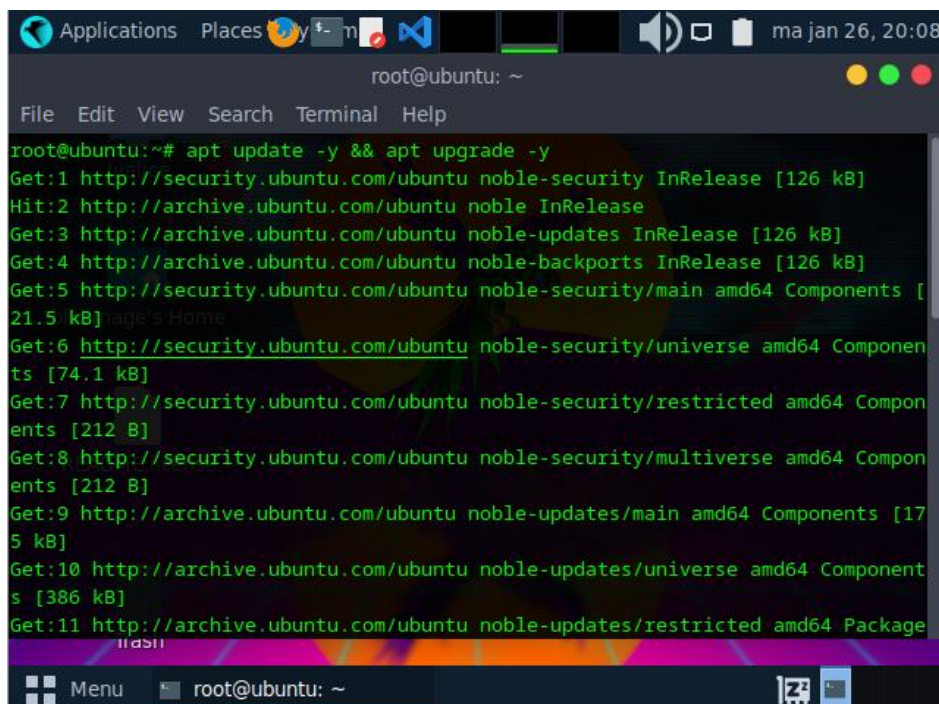
► Instalar Apache Tomcat y verificar su funcionamiento.

Conectamos a nuestro dominio:

A terminal window titled 'root@ubuntu: ~' with a menu bar (File, Edit, View, Search, Terminal, Help) and a system bar (Applications, Places, icons, ma jan 26, 20:06). The terminal shows an SSH session to 'biskmatar.com'. It displays a warning about host keys, a fingerprint, and a confirmation to continue. After entering the password, it shows the Ubuntu 24.04.3 LTS login banner and links for documentation, management, and support.

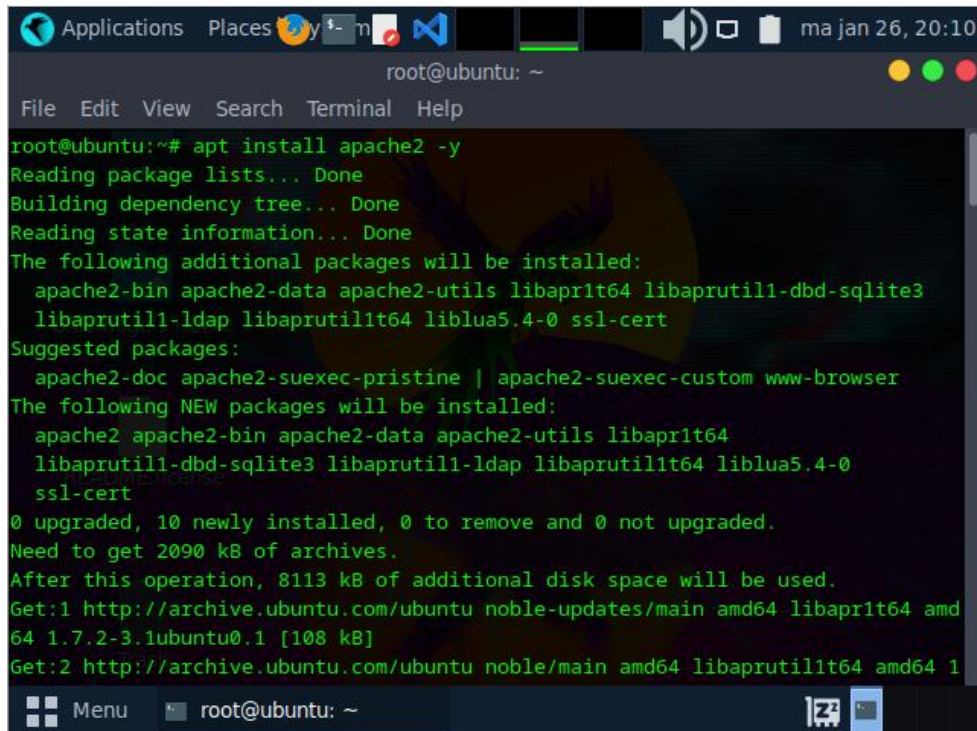
```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu: ~ 20:05:21  
ssh root@biskmatar.com  
hostkeys_find_by_key_hostfile: hostkeys_foreach failed for /home/bluemage/.ssh/  
/known_hosts: Permission denied  
The authenticity of host 'biskmatar.com (217.154.102.175)' can't be establishe  
d.  
ED25519 key fingerprint is SHA256:eYk5SYr6+Hy9p+UuSIMKG6grZa0EX3TE/+rjeSEQ1IA.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Failed to add the host to the list of known hosts (/home/bluemage/.ssh/known_h  
osts).  
root@biskmatar.com's password:  
client_input_hostkeys: hostkeys_foreach failed for /home/bluemage/.ssh/known_h  
osts: Permission denied  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro
```

Actualizamos:

A terminal window titled 'root@ubuntu: ~' with a menu bar (File, Edit, View, Search, Terminal, Help) and a system bar (Applications, Places, icons, ma jan 26, 20:08). The terminal shows the execution of 'apt update -y' and 'apt upgrade -y'. It lists various package sources and their update sizes, including security updates and main components.

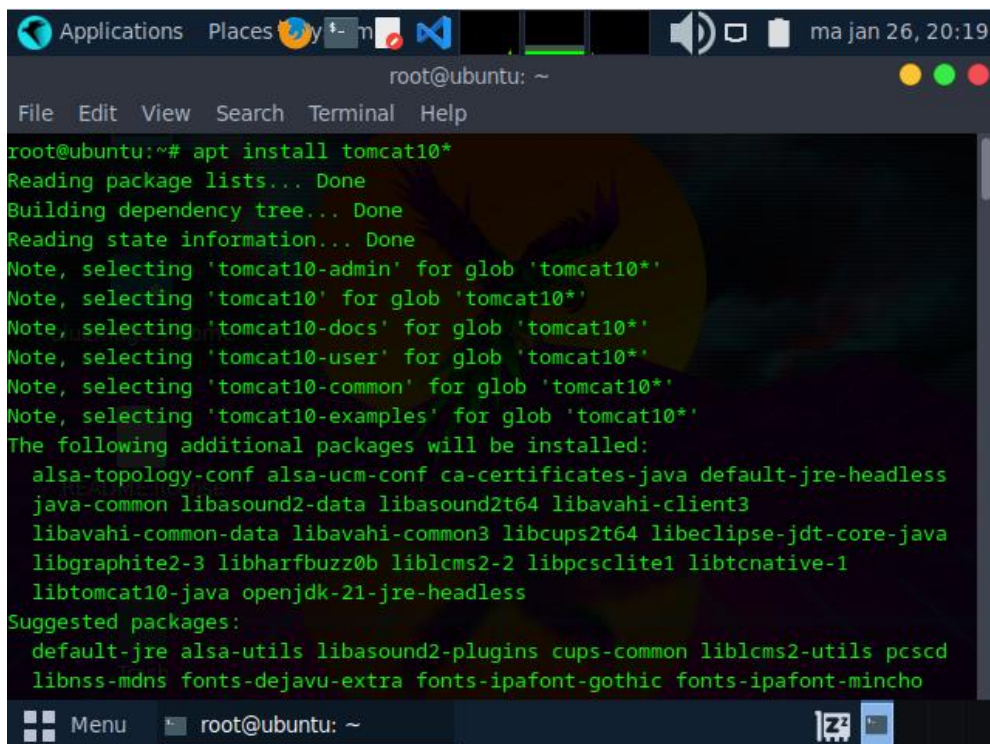
```
root@ubuntu: ~# apt update -y && apt upgrade -y  
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Component  
s [74.1 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Compon  
ents [212 B]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Compon  
ents [212 B]  
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]  
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Component  
s [386 kB]  
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Package
```

Instalamos apache:



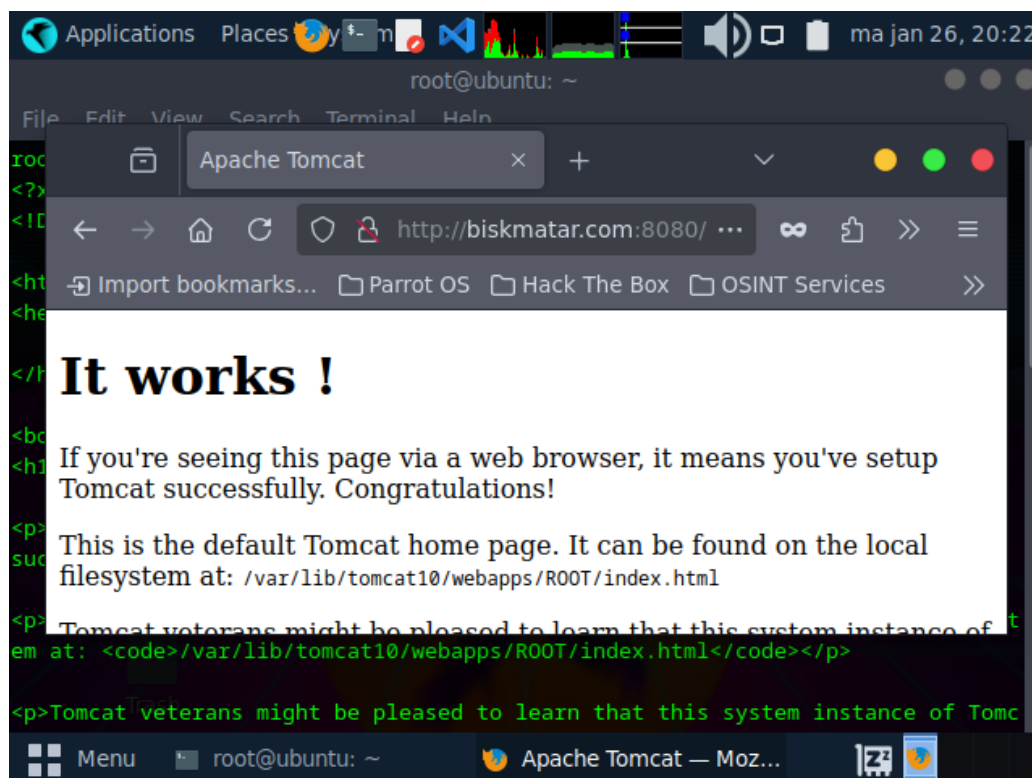
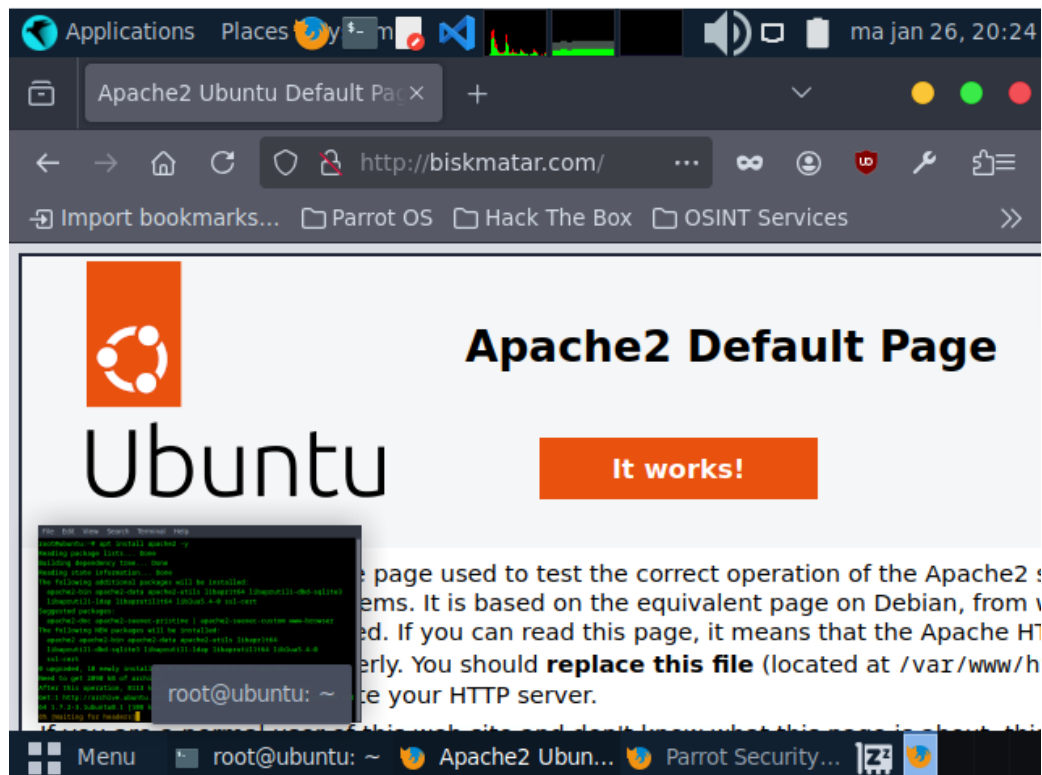
```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# apt install apache2 -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3  
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert  
Suggested packages:  
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser  
The following NEW packages will be installed:  
  apache2 apache2-bin apache2-data apache2-utils libapr1t64  
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0  
  ssl-cert  
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.  
Need to get 2090 kB of archives.  
After this operation, 8113 kB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd  
64 1.7.2-3.1ubuntu0.1 [108 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1
```

Instalamos tomcat:

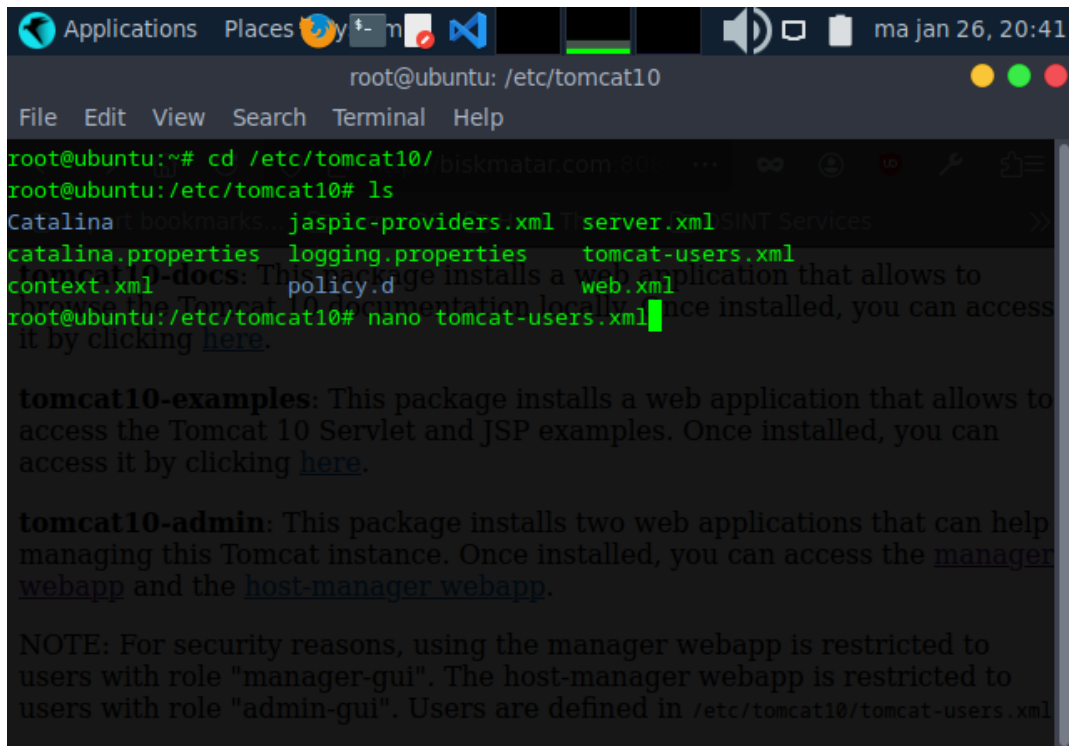


```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# apt install tomcat10*  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Note, selecting 'tomcat10-admin' for glob 'tomcat10*'  
Note, selecting 'tomcat10' for glob 'tomcat10*'  
Note, selecting 'tomcat10-docs' for glob 'tomcat10*'  
Note, selecting 'tomcat10-user' for glob 'tomcat10*'  
Note, selecting 'tomcat10-common' for glob 'tomcat10*'  
Note, selecting 'tomcat10-examples' for glob 'tomcat10*'  
The following additional packages will be installed:  
  alsa-topology-conf alsa-ucm-conf ca-certificates-java default-jre-headless  
  java-common libasound2-data libasound2t64 libavahi-client3  
  libavahi-common-data libavahi-common3 libcups2t64 libecjclipse-jdt-core-java  
  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1  
  libtomcat10-java openjdk-21-jre-headless  
Suggested packages:  
  default-jre alsa-utils libasound2-plugins cups-common liblcms2-utils pcsd  
  libnss-midns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho
```

Comprobamos que funciona:



Creamos usuario para acceder al manager:



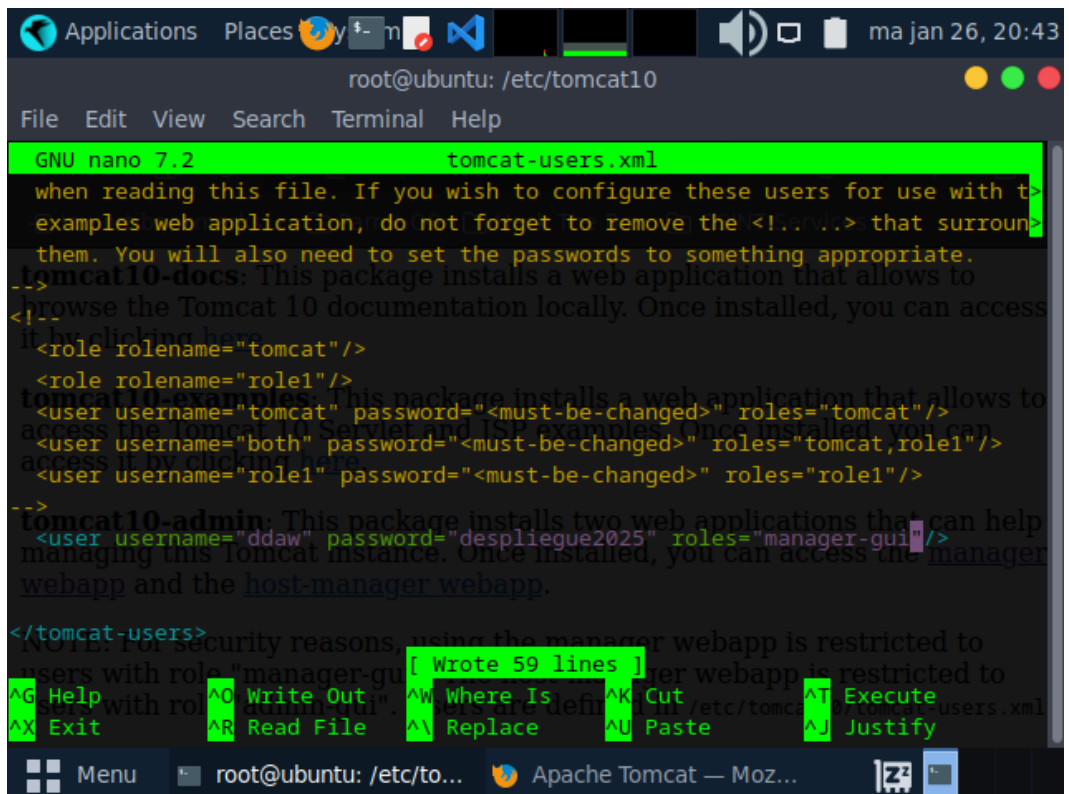
```
root@ubuntu: /etc/tomcat10
File Edit View Search Terminal Help
root@ubuntu:~# cd /etc/tomcat10/
root@ubuntu:/etc/tomcat10# ls
Catalina  jaspic-providers.xml  server.xml  SINT Services
catalina.properties  logging.properties  tomcat-users.xml
context.xml  policy.d  web.xml
root@ubuntu:/etc/tomcat10# nano tomcat-users.xml
```

tomcat10-docs: This package installs a web application that allows to browse the Tomcat 10 documentation locally. Once installed, you can access it by clicking [here](#).

tomcat10-examples: This package installs a web application that allows to access the Tomcat 10 Servlet and JSP examples. Once installed, you can access it by clicking [here](#).

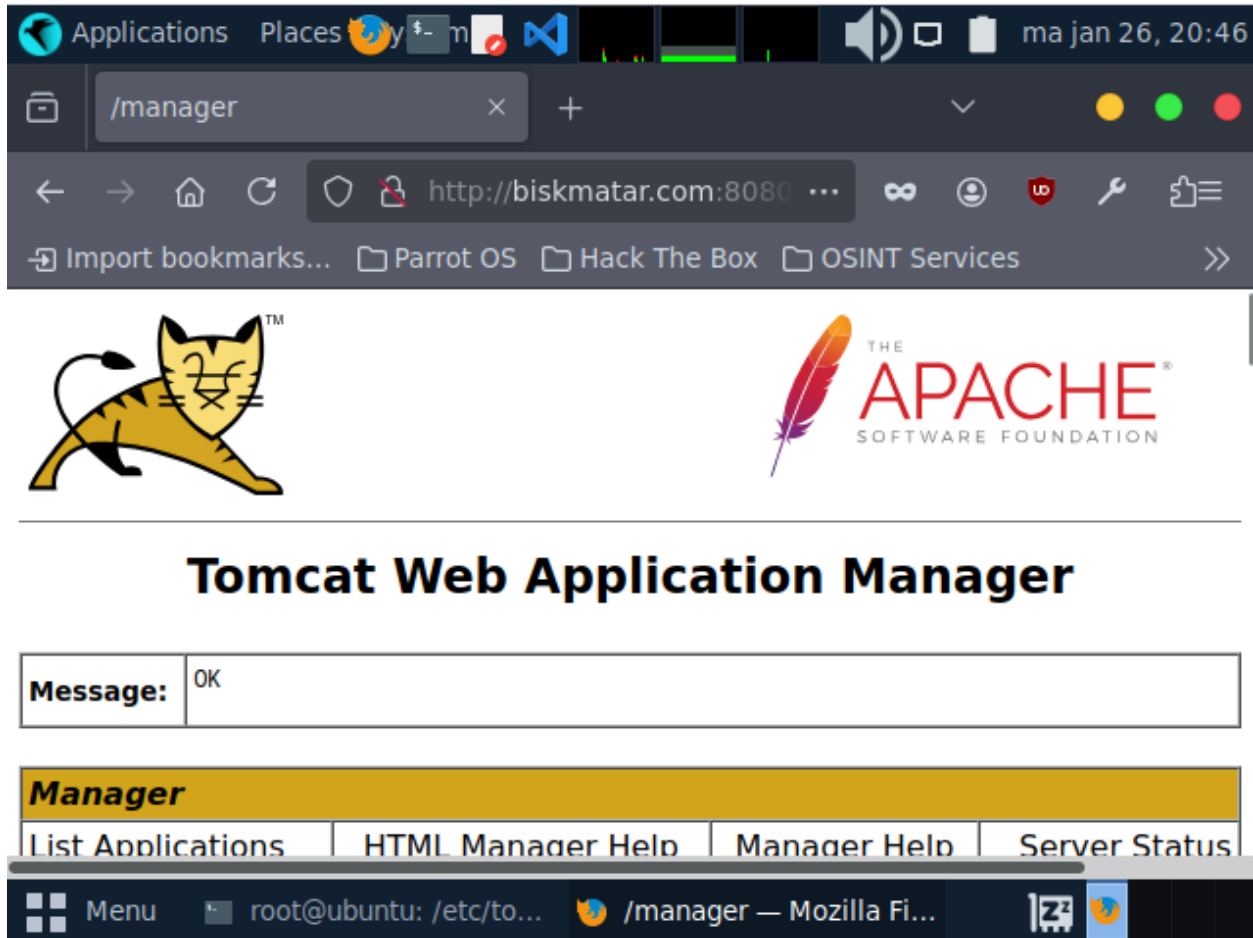
tomcat10-admin: This package installs two web applications that can help managing this Tomcat instance. Once installed, you can access the [manager webapp](#) and the [host-manager webapp](#).

NOTE: For security reasons, using the manager webapp is restricted to users with role "manager-gui". The host-manager webapp is restricted to users with role "admin-gui". Users are defined in /etc/tomcat10/tomcat-users.xml



```
GNU nano 7.2 tomcat-users.xml
when reading this file. If you wish to configure these users for use with t>
examples web application, do not forget to remove the <!-- --> that surroun>
them. You will also need to set the passwords to something appropriate.
tomcat10-docs: This package installs a web application that allows to
browse the Tomcat 10 documentation locally. Once installed, you can access
it by clicking here.
<!--
<role rolename="tomcat"/>
<role rolename="role1"/>
<user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
tomcat10-admin: This package installs two web applications that can help
managing this Tomcat instance. Once installed, you can access the manager
webapp and the host-manager webapp.
NOTE: For security reasons, using the manager webapp is restricted to
users with role "manager-gui". The host-manager webapp is restricted to
users with role "admin-gui". Users are defined in /etc/tomcat10/tomcat-users.xml
[ Wrote 59 lines ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
Menu root@ubuntu: /etc/to... Apache Tomcat — Moz...
```

Aquí estaríamos con acceso al manager:



The screenshot shows a web browser window with the address bar displaying `http://biskmatar.com:8080`. The page features the Tomcat logo (a yellow cat) and the Apache Software Foundation logo. The main heading is "Tomcat Web Application Manager". Below this, there is a "Message:" field with the text "OK". A yellow bar labeled "Manager" is visible, followed by a navigation menu with links: "List Applications", "HTML Manager Help", "Manager Help", and "Server Status". The browser's status bar at the bottom shows the user is logged in as "root@ubuntu" and the current page is "/manager".

Applications Places ma jan 26, 20:46

Import bookmarks... Parrot OS Hack The Box OSINT Services >>

Tomcat Web Application Manager

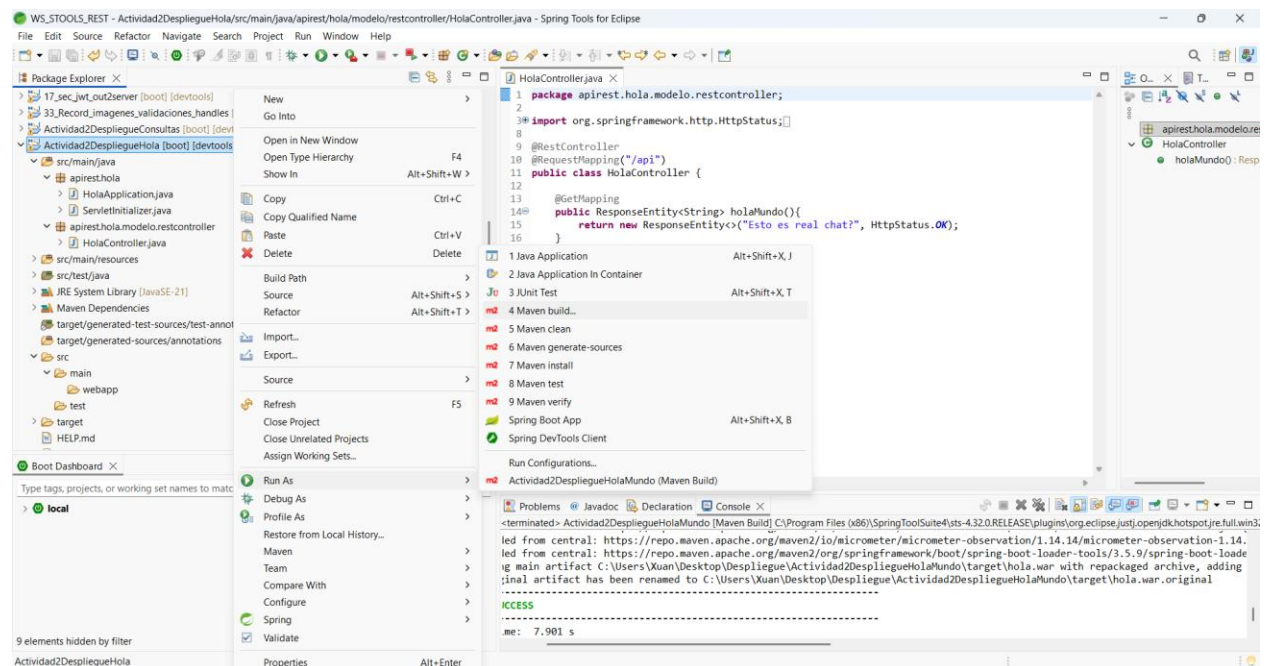
Message: OK

Manager

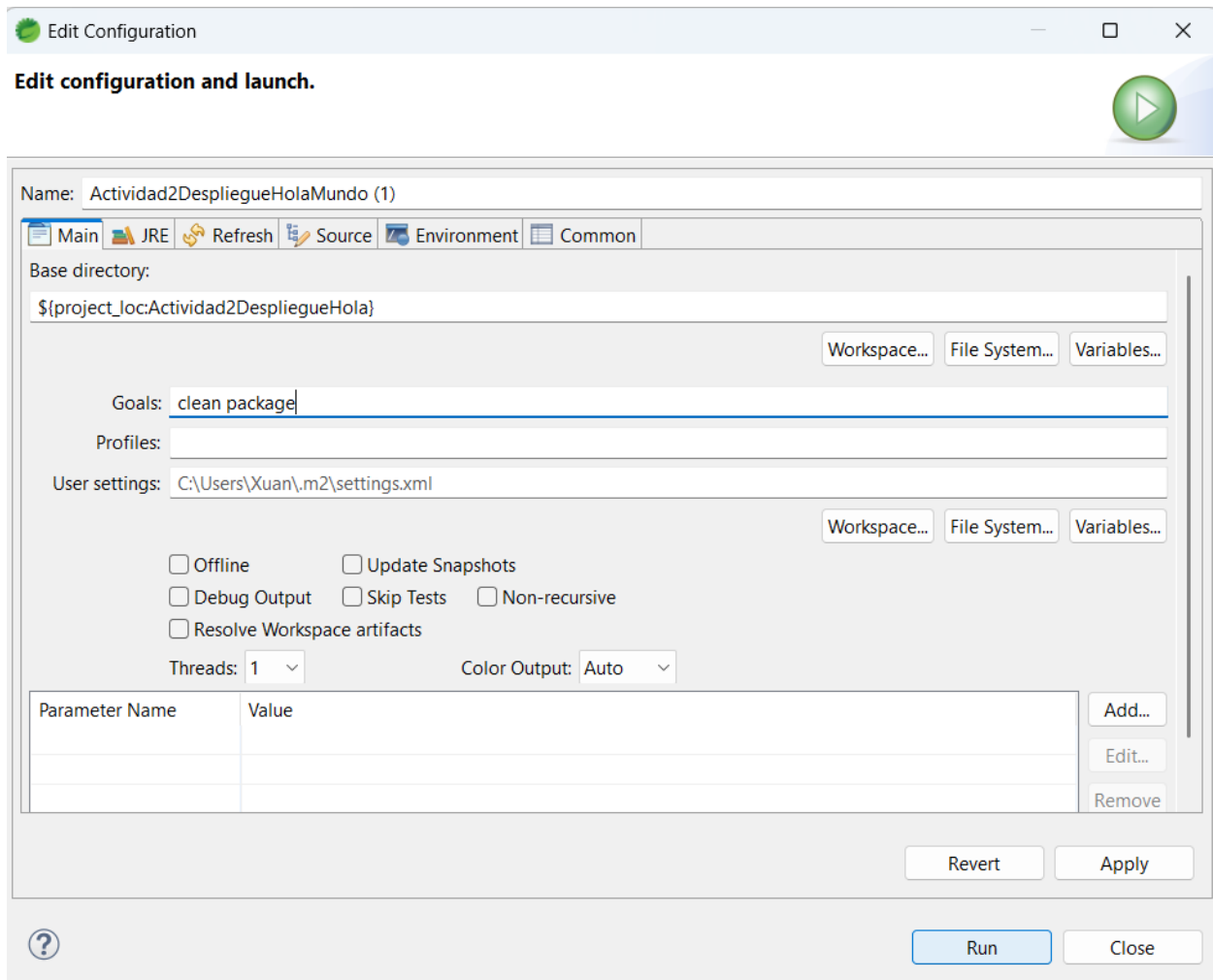
List Applications HTML Manager Help Manager Help Server Status

Menu root@ubuntu: /etc/to... /manager — Mozilla Fi...

- Creamos el .war con SpringToolSuite4, boton derecho en el proyecto y vamos a Maven build...:

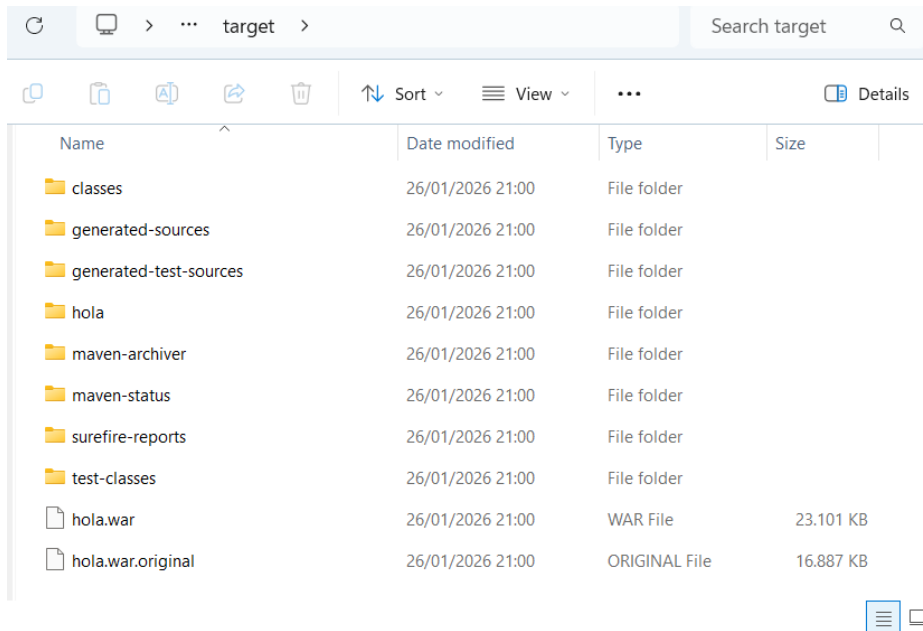


Introducimos “clean package” en Goals y pulsamos run:



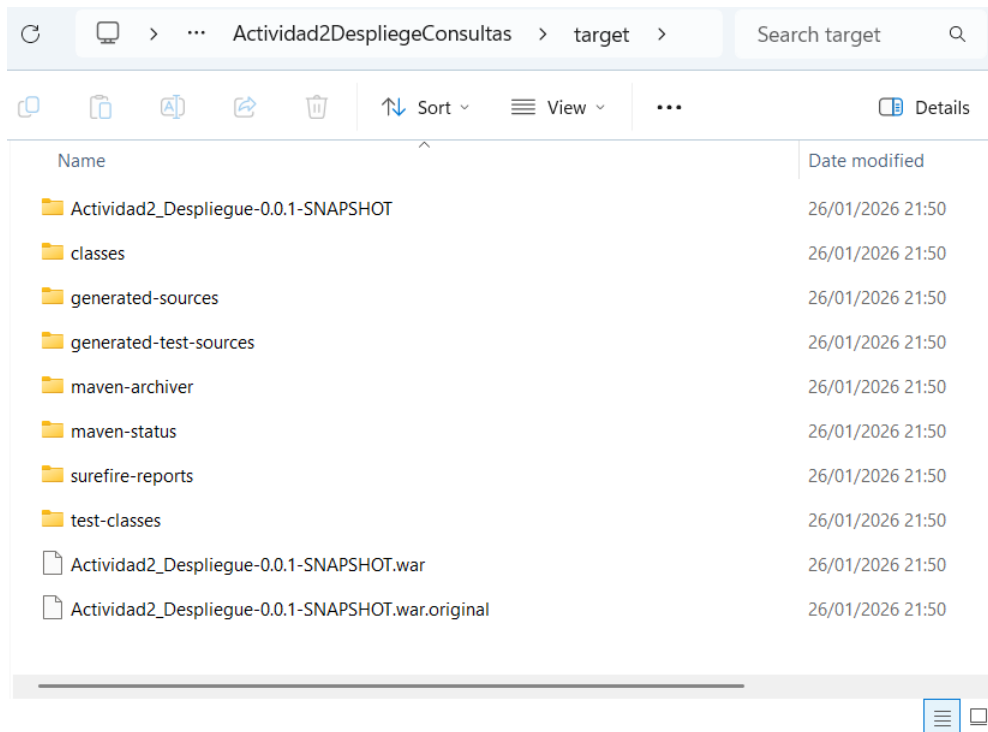
Para el proyecto del CRUD serían los mismos pasos

Comprobamos que tenemos ambos archivos .war en los directorios “target” de los proyectos:



Search target

Name	Date modified	Type	Size
classes	26/01/2026 21:00	File folder	
generated-sources	26/01/2026 21:00	File folder	
generated-test-sources	26/01/2026 21:00	File folder	
hola	26/01/2026 21:00	File folder	
maven-archiver	26/01/2026 21:00	File folder	
maven-status	26/01/2026 21:00	File folder	
surefire-reports	26/01/2026 21:00	File folder	
test-classes	26/01/2026 21:00	File folder	
hola.war	26/01/2026 21:00	WAR File	23.101 KB
hola.war.original	26/01/2026 21:00	ORIGINAL File	16.887 KB



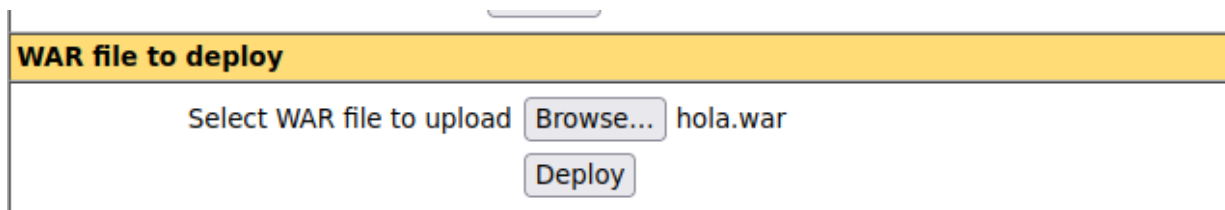
Search target

Name	Date modified
Actividad2_Despliegue-0.0.1-SNAPSHOT	26/01/2026 21:50
classes	26/01/2026 21:50
generated-sources	26/01/2026 21:50
generated-test-sources	26/01/2026 21:50
maven-archiver	26/01/2026 21:50
maven-status	26/01/2026 21:50
surefire-reports	26/01/2026 21:50
test-classes	26/01/2026 21:50
Actividad2_Despliegue-0.0.1-SNAPSHOT.war	26/01/2026 21:50
Actividad2_Despliegue-0.0.1-SNAPSHOT.war.original	26/01/2026 21:50

► Desplegar los archivos .war en el directorio webapps de Tomcat.

► Probar ambos proyectos en el navegador.

En la página del Application Manager de Tomcat subimos el hola.war:



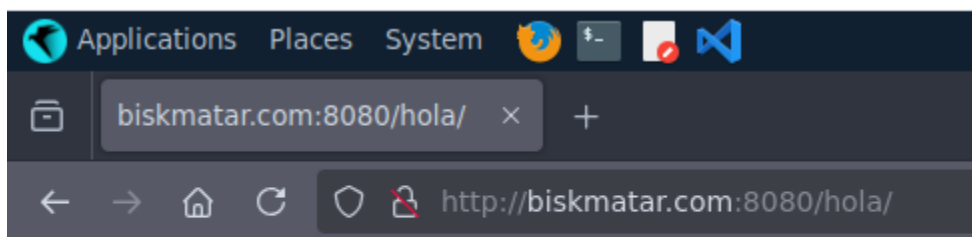
WAR file to deploy

Select WAR file to upload hola.war

Nos aparece en applications:

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/hola	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Clickamos en /hola y:



Esto es real chat?

Nuestro archivo “Actividad2Despliegue” ocupa mas de 50mbs por lo que no Podemos subirlo por el anterior método así que hacemos lo siguiente:

Subimos el archivo:

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> scp C:\Users\Xuan\Desktop\Despliegue\Actividad2Despliegue.war root@biskmatar.com:/tmp/
root@biskmatar.com's password:
Actividad2Despliegue.war
PS C:\WINDOWS\system32> 100% 52MB 27.9MB/s 00:01
```

Nos conectamos al dominio:

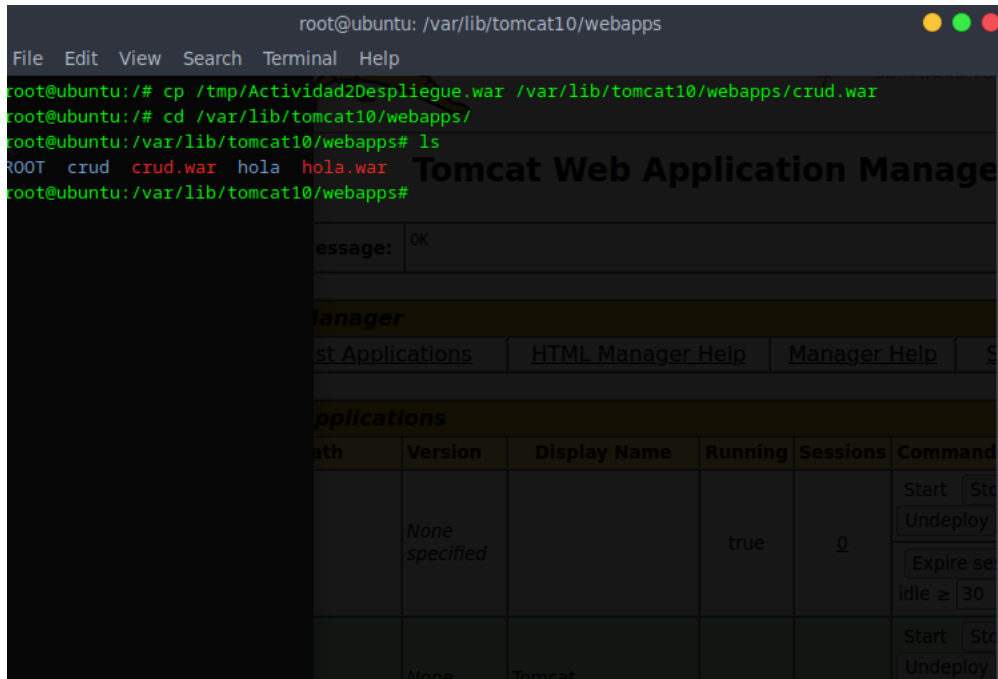
```
root@ubuntu: ~
File Edit View Search Terminal Help
~ > ssh root@biskmatar.com
Warning: Permanently added 'biskmatar.com' (217.154.102.175) to the list of known hosts.
Permission denied (publickey).
The authenticity of host 'biskmatar.com (217.154.102.175)' can't be established.
D25519 key fingerprint is SHA256:eYk5SYr6+Hy9p+UuSIMKG6grZa0EX3TE/+rjeSEQ11A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Failed to add the host to the list of known hosts (/home/blueimage/.ssh/known_hosts).
root@biskmatar.com's password:anager
Client input hostkeys: hostkeys_foreach failed for /home/blueimage/.ssh/known_hosts: Permission denied
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jan 26 21:40:30 UTC 2026

System load: 0.02          Processes:            105
Usage of /: 25.5% of 8.65GB Users logged in:        1
Memory usage: 53%          IPv4 address for ens6: 217.154.102.175
Swap usage: 0%
```

Lo movemos y lo cambiamos a crud.war para mas agilidad:



Moviendolo ahí ya nos aparecería:

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/crud	None specified		false	0	Start Stop Reload Undeploy
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/hola	None specified		true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Nos aparece el siguiente problema:

Message:	FAIL - Application at context path [/crud] could not be started FAIL - Encountered exception [org.apache.catalina.LifecycleException: Failed to start component [StandardEngine[Catalina].StandardHost[localhost].StandardContext[/crud]]]
----------	---

Esto es a causa de que falta la base de datos

Instalamos mariadb y lo activamos:

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
root@ubuntu:/var/lib/tomcat10/webapps# apt install mariadb-server mariadb-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  galera-4 libbcbi-fast-perl libbcbi-pm-perl libclone-perl libconfig-inifiles-perl
  libdbd-mysql-perl libdbi-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3 libmysqlclient21
  libsnappy1v5 libtimedate-perl liburi-perl liburing2 mariadb-client-core mariadb-common
  mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4 mariadb-plugin-provider-lzma
  mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy mariadb-server-core
  mysql-common pv socat
Suggested packages:
  libldb-perl libnet-daemon-perl libsql-statement-perl libdata-dump-perl
  libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl
  libregexp-ipv6-perl libwww-perl mailx mariadb-test doc-base
The following NEW packages will be installed:
  galera-4 libbcbi-fast-perl libbcbi-pm-perl libclone-perl libconfig-inifiles-perl
  libdbd-mysql-perl libdbi-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3 libmysqlclient21
  libsnappy1v5 libtimedate-perl liburi-perl liburing2 mariadb-client mariadb-client-core
  mariadb-common mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4
```

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
root@ubuntu:/var/lib/tomcat10/webapps# systemctl start mariadb
root@ubuntu:/var/lib/tomcat10/webapps# systemctl enable mariadb
Synchronizing state of mariadb.service with SysV service script with /usr/lib/sy
stemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mariadb
root@ubuntu:/var/lib/tomcat10/webapps# systemctl status mariadb
• mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: e
   nabled)
   Active: active (running) since Mon 2026-01-26 22:02:06 UTC; 2min 15s ago
   Docs: man:mariabdb(8)
         https://mariadb.com/kb/en/library/systemd/
   Main PID: 6560 (mariabdb)
   CGroup: /system.slice/mariadb.service
           └─6560 /usr/sbin/mariabdb false

Jan 26 22:02:05 ubuntu mariabdb[6560]: 2026-01-26 22:02:05 0 [Note] InnoDB: log
```

Entramos a mariadb e introducimos la query de sql:

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help

root@ubuntu:/var/lib/tomcat10/webapps# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04
Option: Failed to start component [StandardEngine[Catalina].StandardHost[localhost].StandardEngine
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

HTML Manager Help Manage
MariaDB [(none)]> CREATE DATABASE GESTION_PEDIDOS_BBDD;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> USE GESTION_PEDIDOS_BBDD;
Database changed
MariaDB [GESTION_PEDIDOS_BBDD]> CREATE TABLE USUARIOS
-> ( ID_USUARIO INT auto_increment PRIMARY KEY,
-> NOMBRE VARCHAR(50) NOT NULL,
-> EMAIL VARCHAR(50) NOT NULL UNIQUE,
-> PASSWORD VARCHAR(45) NOT NULL,
-> ENABLED INT,
```


La app debería funcionar pero nos da el siguiente error:

```
MariaDB [GESTION_PEDIDOS_BBDD]> COMMIT; true 0
Query OK, 0 rows affected (0.000 sec)

MariaDB [GESTION_PEDIDOS_BBDD]>
MariaDB [GESTION_PEDIDOS_BBDD]> ^C
MariaDB [GESTION_PEDIDOS_BBDD]> exit
Bye
root@ubuntu:/var/lib/tomcat10/webapps# systemctl status tomcat10
* tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset:→ Reload (
   Active: failed (Result: oom-kill) since Mon 2026-01-26 22:02:35 UTC; 8min>
   Duration: 2h 17min 20.578s
   Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Main PID: 4148 (code=killed, signal=KILL)
   CPU: 1min 30.026s

Jan 26 21:51:27 ubuntu tomcat10[4148]: at org.hibernate.engine.jdbc.env>
Jan 26 21:51:27 ubuntu tomcat10[4148]: at org.hibernate.engine.jdbc.env>
Jan 26 21:51:27 ubuntu tomcat10[4148]: at org.hibernate.engine.jdbc.env>
Jan 26 21:51:27 ubuntu tomcat10[4148]: at org.hibernate.boot.registry.i>
Jan 26 21:51:27 ubuntu tomcat10[4148]: at org.hibernate.service.interna>
Jan 26 21:51:27 ubuntu tomcat10[4148]: ... 66 common frames omitted
Jan 26 22:02:35 ubuntu systemd[1]: tomcat10.service: A process of this unit has>
Jan 26 22:02:35 ubuntu systemd[1]: tomcat10.service: Main process exited, code=>
Jan 26 22:02:35 ubuntu systemd[1]: tomcat10.service: Failed with result 'oom-ki>
Jan 26 22:02:35 ubuntu systemd[1]: tomcat10.service: Consumed 1min 30.026s CPU >
ct WAR file to upload [Browse...] No file selected.

root@ubuntu:/var/lib/tomcat10/webapps# nano /usr/lib/systemd/system/tomcat10.service
root@ubuntu:/var/lib/tomcat10/webapps# systemctl daemon-reload
root@ubuntu:/var/lib/tomcat10/webapps# systemctl start tomcat10
root@ubuntu:/var/lib/tomcat10/webapps# free -h
© 23:16:56
```

Que preguntando a la ia significa lo siguiente:

Diagnosed memory exhaustion and outlined remediation steps. 

ERROR: OOM-KILL = Out Of Memory (Sin memoria RAM)

El WAR del CRUD (51 MB + Spring Boot) consumió toda la RAM del servidor y el sistema lo mató.

SOLUCIÓN INMEDIATA: Eliminar el WAR problemático

```
bash

# Eliminar el WAR y su directorio desplegado
rm /var/lib/tomcat10/webapps/crud.war
rm -rf /var/lib/tomcat10/webapps/crud/

# Verificar cuánta RAM tiene el servidor
free -h

# Reiniciar Tomcat
systemctl start tomcat10
systemctl status tomcat10
```

PROBLEMA: Tu servidor tiene poca RAM

Spring Boot es pesado. Necesitas:

Opción A: Aumentar RAM del VPS (en IONOS)

- Actualiza el plan a uno con más RAM (mí  2GB recomendado)

No temenos suficiente RAM en nuestro server

► Instalar el paquete vsftpd

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
root@ubuntu:/var/lib/tomcat10/webapps# apt install vsftpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  vsftpd
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 120 kB of archives.
After this operation, 312 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu3.1 [120 kB]
Fetched 120 kB in 0s (378 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database ... 78772 files and directories currently installed.)
Preparing to unpack .../vsftpd_3.0.5-0ubuntu3.1_amd64.deb ...
Unpacking vsftpd (3.0.5-0ubuntu3.1) ...
Setting up vsftpd (3.0.5-0ubuntu3.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/systemd/system/vsftpd.service.
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
```

► Editar el archivo de configuración con los parámetros adecuados

Hacemos un backup:

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
root@ubuntu:/var/lib/tomcat10/webapps# cp /etc/vsftpd.conf /etc/vsftpd.conf.backup
root@ubuntu:/var/lib/tomcat10/webapps#
```

Modificamos la configuracion:

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
GNU nano 7.2 /etc/vsftpd.conf *
chroot_local_user=YES
secure_chroot_dir=/var/run/vsftpd/empty
x11_service_name=vsftpd
ssl_enable=NO
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=50000
allow_writeable_chroot=YES
user_sub_token=$USER
local_root=/home/$USER/ftp
t: Web Application Manager
tion: Failed to start component [StandardEngine[Catalina].StandardHost[localhost].StandardContext[/
HTML Manager Help Manager Help
Play Name Running Sessions Commands
G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^G Location
X Exit ^R Read File ^N Replace ^U Paste ^J Justify ^_ Go To Line
```

Creamos directorio ftp, damos permisos y creamos archivo de prueba:

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
root@ubuntu:/var/lib/tomcat10/webapps# mkdir -p /root/ftp/upload
root@ubuntu:/var/lib/tomcat10/webapps# chmod 550 /root/ftp
root@ubuntu:/var/lib/tomcat10/webapps# chmod 750 /root/ftp/upload
root@ubuntu:/var/lib/tomcat10/webapps# echo "Transferencia FTP exitosa - Actividad Despliegue" > /root/ftp/test.txt
root@ubuntu:/var/lib/tomcat10/webapps#
```

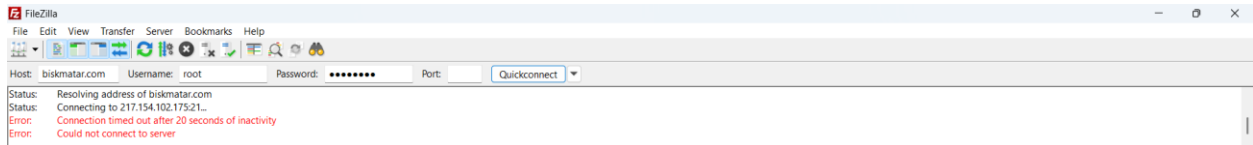
Arrancamos vsftpd:

```
root@ubuntu: /var/lib/tomcat10/webapps
File Edit View Search Terminal Help
root@ubuntu:/var/lib/tomcat10/webapps# systemctl restart vsftpd
root@ubuntu:/var/lib/tomcat10/webapps# systemctl enable vsftpd
synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
root@ubuntu:/var/lib/tomcat10/webapps# systemctl status vsftpd
• vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-01-26 22:32:26 UTC; 18s ago
   Main PID: 7693 (vsftpd)
     Tasks: 1 (limit: 972)
   Memory: 1.0M (peak: 1.1M)
   CPU: 8ms
   CGroup: /system.slice/vsftpd.service
           └─7693 /usr/sbin/vsftpd /etc/vsftpd.conf

Jan 26 22:32:26 ubuntu systemd[1]: Starting vsftpd.service - vsftpd FTP server.
Jan 26 22:32:26 ubuntu systemd[1]: Started vsftpd.service - vsftpd FTP server.
root@ubuntu:/var/lib/tomcat10/webapps#
HTML Manager Help Manager Help
Play Name Running Sessions Commands
Start Stop Reload Undeploy
```

► Verificar funcionamiento con una transferencia de un fichero a través del filezilla

Nos intentamos conectar pero nos da problema:



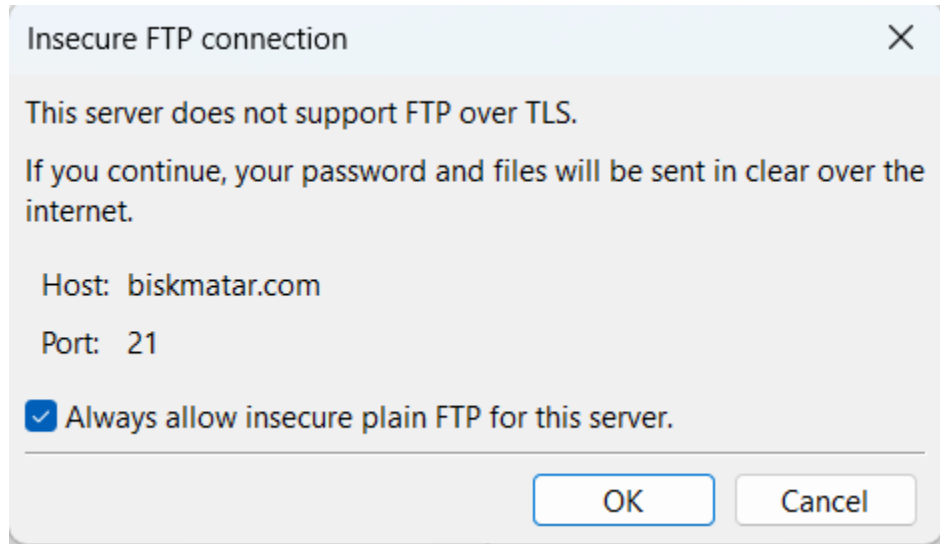
Vamos a IONOS a abrir el puerto 21:

Configuración

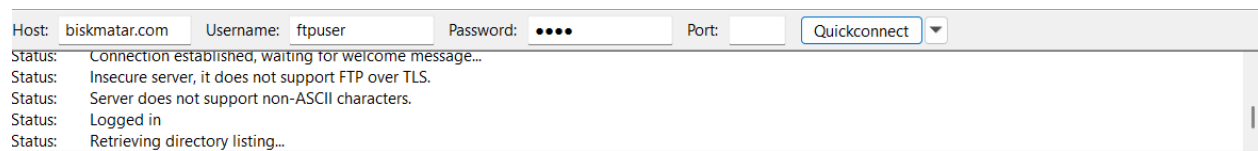
Entrada

Acción▼	IP permitida▼	Protocolo▼	Puerto(s)▼	Descripción▼		
Permitir	Todas	TCP	22			
Permitir	Todas	TCP	80			
Permitir	Todas	TCP	443			
Permitir	Todas	TCP	8443			
Permitir	Todas	TCP	8447			
Permitir	Todas	UDP	53	DNS		
Permitir	Todas	TCP	8080			
Permitir	todas	TCP	21	FTP		

Mensaje de seguridad, damos ok:



Ya estaríamos conectados:



Y así podríamos descargar el archivo:

